RAJALAKSHMI ENGINEERING COLLEGE

Hari Ganesh M 251801081-2025-AIDS

H2

Started on	Sunday, 30 August 2026, 10:26 PM
State	Finished
Completed on	Sunday, 30 August 2026, 10:28 PM
Time taken	1 min 41 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int findFirstZero(int arr[], int low, int high) {
    if (high >= low) {
        int mid = low + (high - low) / 2;


        if ((mid == 0 || arr[mid - 1] == 1) && arr[mid] == 0) {
            return mid;
        }


        if (arr[mid] == 1) {
            return findFirstZero(arr, mid + 1, high);
        }


        return findFirstZero(arr, low, mid - 1);
    }

    return -1;
}

int countZeros(int arr[], int n) {
    int firstZeroIndex = findFirstZero(arr, 0, n - 1);

    if (firstZeroIndex == -1) {
        return 0;
    }

    return n - firstZeroIndex;
}

int main() {
    int m;
    if (scanf("%d", &m) != 1) return 0;

    int *arr = (int *)malloc(m * sizeof(int));
    for (int i = 0; i < m; i++) {
        scanf("%d", &arr[i]);
    }

    printf("%d\n", countZeros(arr, m));

    free(arr);
    return 0;
}
```

	Input	Expected	Got	
✔	5 1 1 1 0 0	2	2	✔
✔	10 1 1 1 1 1 1 1 1 1 1	0	0	✔
✔	8 0 0 0 0 0 0 0 0	8	8	✔
✔	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Hari Ganesh M 251801081-2025-AIDS

H2

Started on	Sunday, 30 August 2026, 10:28 PM
State	Finished
Completed on	Sunday, 30 August 2026, 10:29 PM
Time taken	45 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

Example 1:

```
Input: nums = [3,2,3]
Output: 3
```

Example 2:

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 10^4`
- `-2^31 <= nums[i] <= 2^31 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>


int findMajorityElement(int *nums, int n) {
    int candidate = 0;
    int count = 0;

    for (int i = 0; i < n; i++) {
        if (count == 0) {
            candidate = nums[i];
            count = 1;
        } else if (nums[i] == candidate) {
            count++;
        } else {
            count--;
        }
    }

    return candidate;
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

```

```
27      int *nums = (int *)malloc(n * sizeof(int));
28      for (int i = 0; i < n; i++) {
29          scanf("%d", &nums[i]);
30      }
31
32      printf("%d\n", findMajorityElement(nums, n));
33
34      free(nums);
35      return 0;
36  }
```

	Input	Expected	Got	
✔	3 3 2 3	3	3	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Hari Ganesh M 251801081-2025-AIDS

H2

Started on	Sunday, 30 August 2026, 10:30 PM
State	Finished
Completed on	Sunday, 30 August 2026, 10:34 PM
Time taken	4 mins 37 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>


int findFloor(int arr[], int n, int x) {
    int low = 0, high = n - 1;
    int floorIndex = -1;

    while (low <= high) {
        int mid = low + (high - low) / 2;

        if (arr[mid] == x) {
            return mid;
        }

        if (arr[mid] < x) {
            floorIndex = mid;
            low = mid + 1;
        } else {
            high = mid - 1;
        }
    }

    return floorIndex;
}

int main() {
    int n, x;
    if (scanf("%d", &n) != 1) return 0;

    int *arr = (int *)malloc(n * sizeof(int));
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    if (scanf("%d", &x) != 1) {
        free(arr);
        return 0;
    }

    int index = findFloor(arr, n, x);

    if (index != -1) {
        printf("%d\n", arr[index]); // Prints the floor element (use printf("%d\n", index); if index is
    } else {
        printf("-1\n");
    }

    free(arr);
    return 0;
}
```

```
31 }
```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔
✔	5 10 22 85 108 129 100	85	85	✔
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Hari Ganesh M 251801081-2025-AIDS

H2

Started on	Sunday, 30 August 2026, 10:36 PM
State	Finished
Completed on	Wednesday, 30 September 2026, 3:14 PM
Time taken	30 days 16 hours
Marks	0.00/1.00
Grade	**0.00** out of 10.00 (**0**%)

Question 1 | Not answered Mark 0.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```
1
```

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